

Runoff

Storm Hydrograph Interpretation

Fig. (1) shows a hydrograph that shows stream discharge (y-axis) against time (x-axis) for a small catchment after a rainfall event.

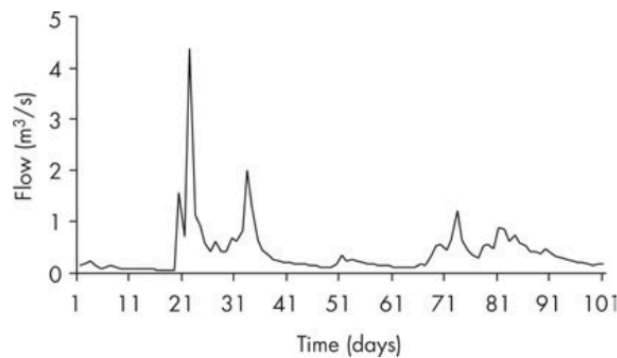


Figure 1: Hydrograph for the river Wye, Wales for a 100-day period during the autumn of 1995.

- Clearly identify and label the **rising limb**, **peak discharge**, and **recession limb** of the hydrograph.
- Explain, in words, the **hydrological processes** responsible for each part:
 - What causes the steep rising limb?
 - Why does the hydrograph peak at a particular time?
 - What processes dominate during the recession limb?
- Discuss how **antecedent soil moisture** and **catchment geology** can alter the hydrograph shape. Based on this discussion can you reason what geology the region has?

Peak Runoff Mechanism Comparison

Hydrologists have proposed different theories for how rainfall becomes streamflow during storm events.

- Compare the following three theories:

- Horton's infiltration-excess overland flow.
- Betson's partial contributing areas.
- Hewlett & Hibbert's saturation overland flow.
- For each theory:
 - Draw a simple sketch of the mechanism.
 - State the **dominant control** (e.g. infiltration capacity, saturated areas).
 - Indicate in which type of environment the theory is most relevant (humid vs. arid, forest vs. urban).
- Explain why the modern **variable source area concept** is considered a synthesis of these views.

Runoff Estimation (Rational Method)

Introduction

The **Rational Method** is widely used for estimating **peak discharge** in small basins (typically <200 acres). It assumes that the maximum runoff occurs when the entire catchment contributes to flow. The Rational method was developed primarily for predicting peak flows and its use is not advised for volume-sensitive routing calculations.

The formula is:

$$Q = C i A$$

where:

- Q = peak runoff
- C = runoff coefficient (dimensionless, 0–1, depends on land use/soil)
- i = rainfall intensity for a duration equal to the **time of concentration** (T_c)
- A = catchment drainage area

This formula is applicable to both US and metric evaluations, as long as consistent units are employed. In traditional US usage, the intensity and area are given in inches-per-hour and acres, respectively. Converting the units leaves a factor of approximately 1.01, which is often neglected, but must be included to match the HydroCAD results.

If a basin has **multiple land uses**, the composite coefficient C_{basin} is found as an **area-weighted average** of the different land-use coefficients.

In order to generate a complete runoff hydrograph, it is assumed that the runoff begins at the start of the storm and increases linearly to the peak value. The peak runoff is sustained until the event duration has elapsed, and then decreases linearly to zero.

Exercise

A 80 acres catchment contains the following land uses:

- 20% residential single-family housing ($C = 0.40$)
- 10% streets and driveways ($C = 0.90$)
- 30% lawn on sandy soil ($C = 0.15$)
- 40% woodland ($C = 0.10$)

A design storm has a rainfall intensity of **4.9 in./hr.** The time of concentration is estimated as 15 minutes.

Tasks:

- Compute the **weighted runoff coefficient** C_{basin} using the area-weighted average.
- Using the Rational Method equation, calculate the **peak discharge** Q in cubic m per second (cms).
- Briefly discuss how the peak runoff would change if the woodland area was converted into parking lots (impervious surface with $C \approx 0.90$).

Groundwater and Subsurface Contributions

Not all storm runoff comes from direct overland flow. Subsurface processes can be equally important.

- Explain the concept of **piston flow** and how it differs from **throughflow** in soils.
- Using the **capillary fringe hypothesis**, describe how groundwater can rapidly contribute to streamflow during a storm.
- Tracer studies often find a significant proportion of “old water” in storm runoff (i.e. water that infiltrated before the event).
 - Why does this occur?

- What does it imply about **flow pathways** and the role of the subsurface in storm response?